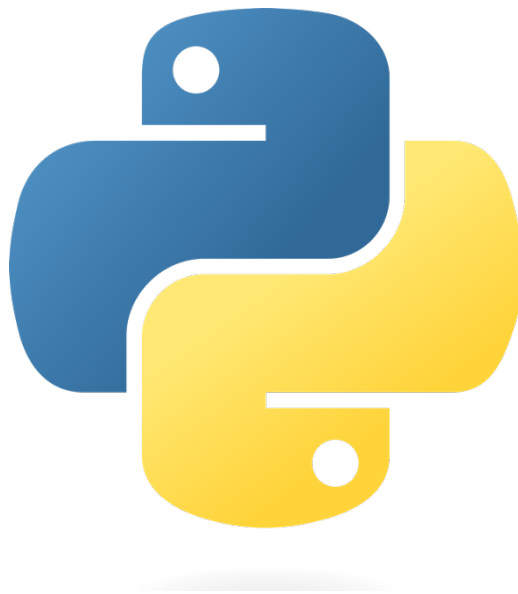


6. File Handling



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1. Python File Open

File handling is an important part of any web application.

Python has several functions for creating, reading, updating, and deleting files.

- **File Handling**

The key function for working with files in Python is the `open()` function.

The `open()` function takes two parameters; *filename*, and *mode*.

There are four different methods (modes) for opening a file:

- `"r"` - Read - Default value. Opens a file for reading, error if the file does not exist
- `"a"` - Append - Opens a file for appending, creates the file if it does not exist
- `"w"` - Write - Opens a file for writing, creates the file if it does not exist
- `"x"` - Create - Creates the specified file, returns an error if the file exists

In addition you can specify if the file should be handled as binary or text mode

- `"t"` - Text - Default value. Text mode
- `"b"` - Binary - Binary mode (e.g. images)

Syntax

To open a file for reading it is enough to specify the name of the file:

```
f = open("demofile.txt")
```

The code above is the same as:

```
f = open("demofile.txt", "rt")
```

Because `"r"` for read, and `"t"` for text are the default values, you do not need to specify them.

Note: Make sure the file exists, or else you will get an error.

2. Python File Open

Open a File on the Server

Assume we have the following file, located in the same folder as Python:

demofile.txt

```
Hello! Welcome to demofile.txt  
This file is for testing purposes.  
Good Luck!
```

To open the file, use the built-in `open()` function.

The `open()` function returns a file object, which has a `read()` method for reading the content of the file:

Example 1

```
f = open("demofile.txt", "r")  
print(f.read())
```

If the file is located in a different location, you will have to specify the file path, like this:

Example 2

Open a file on a different location:

```
f = open("D:\\myfiles\\welcome.txt", "r")  
print(f.read())
```

- **Read Only Parts of the File**

By default the `read()` method returns the whole text, but you can also specify how many characters you want to return:

Example 1

Return the 5 first characters of the file:

```
f = open("demofile.txt", "r")  
print(f.read(5))
```

- **Read Lines**

You can return one line by using the `readline()` method:

Example 1

Read one line of the file:

```
f = open("demofile.txt", "r")
print(f.readline())
```

By calling `readline()` two times, you can read the two first lines:

Example 2

Read two lines of the file:

```
f = open("demofile.txt", "r")
print(f.readline())
print(f.readline())
```

By looping through the lines of the file, you can read the whole file, line by line:

Example 3

Loop through the file line by line:

```
f = open("demofile.txt", "r")
for x in f:
    print(x)
```

- **Close Files**

It is a good practice to always close the file when you are done with it.

Example 1

Close the file when you are finish with it:

```
f = open("demofile.txt", "r")
print(f.readline())
f.close()
```

Note: You should always close your files, in some cases, due to buffering, changes made to a file may not show until you close the file.

3. Python File Write

- **Write to an Existing File**

To write to an existing file, you must add a parameter to the `open()` function:

"a" - Append - will append to the end of the file

"w" - Write - will overwrite any existing content

Example

Open the file "demofile2.txt" and append content to the file:

```
f = open("demofile2.txt", "a")
f.write("Now the file has more content!")
f.close()

#open and read the file after the appending:
f = open("demofile2.txt", "r")
print(f.read())
```

Example

Open the file "demofile3.txt" and overwrite the content:

```
f = open("demofile3.txt", "w")
f.write("Woops! I have deleted the content!")
f.close()

#open and read the file after the overwriting:
f = open("demofile3.txt", "r")
print(f.read())
```

Note: the "w" method will overwrite the entire file.

- **Create a New File**

To create a new file in Python, use the `open()` method, with one of the following parameters:

"x" - Create - will create a file, returns an error if the file exist

"a" - Append - will create a file if the specified file does not exist

"w" - Write - will create a file if the specified file does not exist

Example

Create a file called "myfile.txt":

```
f = open("myfile.txt", "x")
```

Result: a new empty file is created!

Example

Create a new file if it does not exist:

```
f = open("myfile.txt", "w")
```

4. Python Delete File

- **Delete a File**

To delete a file, you must import the OS module, and run its `os.remove()` function:

Example

Remove the file "demofile.txt":

```
import os  
os.remove("demofile.txt")
```

- **Check if File exist:**

To avoid getting an error, you might want to check if the file exists before you try to delete it:

Example

Check if file exists, *then* delete it:

```
import os  
if os.path.exists("demofile.txt"):  
    os.remove("demofile.txt")  
else:  
    print("The file does not exist")
```

- **Delete Folder**

To delete an entire folder, use the `os.rmdir()` method:

Example

Remove the folder "myfolder":

```
import os  
os.rmdir("myfolder")
```

Note: You can only remove *empty* folders.

Reference

1. https://www.w3schools.com/python/python_file_handling.asp
2. https://www.w3schools.com/python/python_file_open.asp
3. https://www.w3schools.com/python/python_file_write.asp
4. https://www.w3schools.com/python/python_file_remove.asp